Improving Quality of Variational Autoencoder Outputs for Ki67 Histological Image Generation

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***Abstract*—Ki67 immunohistochemical staining serves as a proliferation marker in cancer diagnosis, but acquiring large datasets remains challenging. Variational Autoencoders (VAEs) offer a promising approach for generating synthetic Ki67-stained images; however, standard VAE architectures produce blurred outputs, which limit their applicability in medical imaging tasks. This work systematically investigates methods to improve the sharpness of VAE-generated Ki67-stained histopathological images while maintaining stable training dynamics. We evaluate three complementary approaches: (1) alternative reconstruction loss functions, (2) latent space dimensionality scaling, and (3) explicit sharpness regularisation. Our experiments demonstrate that these modifications yield measurable improvements in output sharpness. Our findings show that simple modifications to the standard VAE model can substantially improve output quality without requiring complex adversarial training mechanisms, providing a practical approach for medical image generation or reconstruction in resource-constrained settings.**

***Index Terms*—variational autoencoder, generative models, medical image synthesis, digital pathology, Ki67.**

I. Introduction

Ki67 immunohistochemical staining is a widely used technique in cancer diagnosis that evaluates the expression of the Ki67 protein in tissue sections. Ki67 serves as a cellular proliferation marker, with Ki67-positive cells indicating active cell division and thereby providing an estimate of tumour growth rate [1]. In Ki67-stained images, proliferating (Ki67-positive) cells appear brown, while non-proliferating cells appear blue. However, the Ki67 staining process is resource-intensive, time-consuming, and not always available.

Generative models offer a promising solution by synthesising realistic images that can augment limited training datasets for machine learning applications. Beyond data augmentation, generative models have also been integrated into classification frameworks, as discussed in Section II. Variational Autoencoder (VAE) [2] is a generative model that learns a probabilistic representation of the training data. Architecturally, a VAE comprises two networks: an encoder that projects input data onto a lower-dimensional latent space, and a decoder that reconstructs the data from this compressed space. Specifically, the encoder parameterises the input as a probability distribution, typically a Gaussian, by estimating its mean and variance. The decoder then generates new samples by drawing from this latent distribution. This probabilistic formulation ensures a continuous latent space, enabling the model to generate diverse, novel outputs. The objective function L_{VAE} comprises two terms: a reconstruction loss L_{rec} that ensures the decoder can accurately reconstruct the input, and the Kullback-Leibler (KL) divergence D_{KL} that regularises the latent space to adhere to a prior distribution, typically a standard normal distribution. Formally, the objective function is defined as:

$$L_{VAE} = L_{rec} + D_{KL} \tag{1}$$

While VAEs offer stable training and a probabilistic basis, they tend to produce blurry outputs, which is a known limitation of this architecture [3].

Various generative models have been applied to histological image synthesis, including VAEs, Generative Adversarial Networks (GANs) [4], and more recently, diffusion models [5]. While GANs can produce visually sharp images, they often suffer from training instability and mode collapse. Diffusion models have shown promising results in generating high-quality images but require substantial computational resources and lengthy inference times. In contrast, VAEs offer stable training and computationally efficient sampling; however, standard VAE architectures tend to produce blurred outputs. Generating high-quality histological images with VAEs that preserve sufficient detail and sharpness remains a significant challenge. Blurred outputs may fail to capture fine cellular structures and tissue patterns essential for clinical applications. Addressing this limitation is crucial for enabling the practical deployment of VAE models in medical imaging workflows.

In this work, we investigate methods to improve the sharpness of VAE-generated Ki67-stained histological images. While generative models have been applied to various histopathological imaging tasks, their application to Ki67-stained images remains relatively limited, and addressing the sharpness limitations of standard VAEs in this domain has received little attention. To improve the sharpness of generated images and identify which parameters have the greatest impact on output quality, we conduct three complementary experiments: (1) comparing alternative reconstruction loss functions, (2) evaluating the impact of latent space dimensionality, and

(3) incorporating explicit sharpness regularisation based on the Sobel operator into the training objective. Our experimental results demonstrate measurable improvements in output sharpness, with the final model achieving a significant increase over the baseline model while successfully maintaining the generation of realistic Ki67-stained images.

The paper is organised as follows: Section II discusses related work. Section III describes our dataset, sharpness evaluation methodology, and the VAE configurations under investigation. Section IV presents the experimental results, and Section V concludes with a discussion of findings and directions for future work.

II. Related Work

Several studies have explored VAEs for medical image analysis. Uzunova et al. [6] proposed an unsupervised approach for pathology detection in medical images using a conditional variational autoencoder (CVAE). Their method trains a CVAE on healthy images, leveraging the assumption that pathological regions would result in poor reconstruction quality and map to distant regions in the learned latent space. While demonstrating promising results on brain MRI and thorax CT data, authors acknowledged that VAEs tend to reconstruct images with noticeable blur compared to other generative models, which also motivates our research into improving sharpness in standard VAE-generated medical images.

Guleria et al. [7] explored a hybrid VAE-CNN architecture to enhance diagnostic accuracy in breast histopathology. The VAE served as a generative preprocessing step, and the reconstructed images were then utilised for training a convolutional neural network classifier. Their approach improved classification accuracy; however, researchers did not address the blurriness of reconstructed images.

To address the blurriness inherent to VAE outputs, several hybrid approaches have been proposed. Larsen et al. [3] introduced a model that combines a VAE with a GAN to address output blurriness by replacing element-wise reconstruction errors with feature-wise errors extracted from the GAN discriminator. They demonstrated improved visual fidelity on non-medica image datasets. Similarly, Naderi et al. [8] proposed the Sharpened Adversarial Variational Autoencoder (AVAE-S), which combines a specialised compound loss function with adversarial fine-tuning. Their approach first pre-trains a VAE using a multi-component loss function incorporating pixel-wise reconstruction, edge detection, texture, and structural similarity measures. The pre-trained VAE is then fine-tuned using an adversarial discriminator to enhance output sharpness. Researchers demonstrated substantial improvements on non-medical datasets. While these hybrid VAE-GAN architectures effectively produce sharper outputs, they increase both training complexity and architectural requirements. In contrast, our work investigates simpler approaches that modify the standard VAE model to improve sharpness without requiring adversarial components.

Additionally, our prior work [9]–[11] investigated conditional image generation using StyleGAN to synthesise HE-stained histological images conditioned on Ki67 index. While StyleGAN achieved high visual quality, it required extensive training time. This study explores VAE-based generation as an alternative approach, focusing specifically on Ki67-stained images and evaluating methods to mitigate the characteristic blurriness of VAE outputs while maintaining reasonable training time and stability.

III. Methods

A. Dataset

The original dataset comprises 24 Ki67-stained whole slide images (WSIs) of seminoma, testicular tumour, presented in [12]. The WSIs were originally acquired from the Department of Pathology, Jessenius Medical Faculty of Comenius University and University Hospital Martin, Slovakia. The dataset was partitioned into a training set of 15 WSIs and a test set of 9 WSIs. Due to the high dimensionality of the WSIs, we extracted smaller square patches of 128×128 pixels at the second-highest resolution level. All pixel values are normalised to the range $[0, 1]$. Additionally, we implemented a filtering process to remove patches with low quality or insufficient cellular content, resulting in a dataset of high-quality samples. The final dataset comprises $77\,082$ training patches and $17\,751$ test patches.

B. Sharpness Evaluation

To quantitatively assess the sharpness of the generated images, we employ the Sobel operator, which analyses intensity gradients within images. The evaluation process proceeds as follows: First, each image is converted to grayscale, and then the Sobel operator is applied to compute the horizontal (G_x) and vertical (G_y) intensity gradients. The total gradient magnitude G at each pixel is derived by combining these components:

$$G = \sqrt{G_x^2 + G_y^2} \tag{2}$$

and we calculate the sharpness as the variance of this gradient magnitude matrix [13]:

$$sharpness = Var_G \tag{3}$$

A higher variance indicates greater sharpness, characterised by stronger edges and finer details, whereas a lower variance suggests blurring. For the evaluation of each VAE model, we compute the average sharpness across 1000 randomly generated samples. As a reference, images from the dataset exhibit the average sharpness of 103.76. Our objective is to maximise this sharpness metric in the generated images.

C. VAE models

We investigate several VAE configurations to improve the sharpness and quality of generated Ki67-stained histological images. All models are trained for 50 epochs with a batch size of 64. The models are optimised using the Adam optimiser with a learning rate of 0.001.

1) Baseline VAE: The baseline model is based on a VAE architecture with convolutional layers in both the encoder and decoder. Each convolutional layer applies a 4×4 kernel with a stride of 2 and padding of 1. Both the encoder and decoder contain four convolutional layers, using ReLU activation functions, except for the final decoder layer, which utilises Sigmoid to constrain output values to the $[0, 1]$ range. The latent space has a dimensionality of 128. The model is optimised using the standard VAE objective function (Eq. 1), which comprises two equally weighted components: the reconstruction loss and KL divergence. Reconstruction loss is implemented as mean squared error (MSE):

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(x_i - \hat{x}_i)^2 \tag{4}$$

where x_i represents the original input, $\hat{x}_i$ denotes the reconstructed image, and n is the total number of samples.

2) VAE with Alternative Reconstruction Losses: The reconstruction loss ensures that the decoder output closely resembles the input, thereby learning an effective latent representation. In the baseline architecture, MSE is employed for this objective. To investigate the impact of different reconstruction losses on output quality and sharpness, we scrutinise two alternative loss functions: Mean Absolute Error (MAE) and Binary Cross-Entropy (BCE):

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|x_i - \hat{x}_i| \tag{5}$$

$$BCE = -\frac{1}{n}\sum_{i=1}^{n}\left(x_i \log(\hat{x}_i) + (1 - x_i)\log(1 - \hat{x}_i)\right) \tag{6}$$

Here, similarly to MSE, x_i represents the original image, $\hat{x}_i$ denotes the reconstructed output, and n is the total number of samples. Across the experiments, all other hyperparameters and architectural components remain unchanged.

3) VAE with Alternative Latent Space Dimensionalities: The latent space in the VAE encodes a compressed representation of input data and serves as the basis for generating outputs. Its dimensionality directly governs the model's capacity to capture various features of the input data. A latent space that is too small may limit the model's ability to encode diverse data characteristics, resulting in the loss of fine details. Conversely, an excessively large latent space may lead to overfitting and increase computational complexity during training. To evaluate this trade-off, we compared four latent dimensionalities: 128, 256, 512, and 1024. Based on the results from the previous experiment, MAE was selected as the reconstruction loss function across all configurations. All other hyperparameters remained unchanged.

4) VAE with Sharpness-Aware Objective Function: This experiment investigates the direct integration of image sharpness into the model's optimisation objective. We extend the standard VAE objective function by introducing a sharpness regularisation term derived from the variance of the gradient magnitude, computed via the Sobel operator (refer to Section III-B). In this context, higher variance signifies well-defined edges and high-frequency content, whereas lower variance corresponds to blurred outputs. To encourage the generation of sharper images, we treat the maximisation of gradient variance as a minimisation task within the objective function. Consequently, this term is subtracted from the total loss. The modified objective function extends Eq. 1 as follows:

$$L = L_{MAE} + D_{KL} - \lambda Var_G \tag{7}$$

where L_{MAE} represents MAE reconstruction loss, D_{KL} denotes the KL divergence, and Var_G is the variance of the gradient magnitude matrix. The hyperparameter λ serves as a weighting coefficient to balance reconstruction fidelity against sharpness. To identify the optimal trade-off, we evaluated four configurations: $\lambda \in \{0, 1, 2, 3\}$, where $\lambda = 0$ corresponds to the baseline model without sharpness regularisation. For all VAE variants, the latent space dimensionality is fixed at 512.

IV. Results

The experimental results are summarised in Table I, which evaluates the VAE configurations across three primary criteria: maximising the average image sharpness (as detailed in Section III-B), and minimising the reconstruction MAE and the KL divergence, both computed on the test dataset. Our analysis specifically prioritises image sharpness. Notably, MAE decreases as sharpness improves, indicating higher reconstruction fidelity. However, this gain comes at the cost of increased KL divergence, stemming from an inherent trade-off between reconstruction quality and latent space regularity. Furthermore, KL divergence also rises with increasing latent dimensionality. Qualitative results are provided in Fig. 2 and Fig. 3. Firstly, Fig. 2 compares reconstructions by the top-performing model from each experiment. The first row shows the original input images for reference, with subsequent rows displaying the corresponding model reconstructions. Secondly, Fig. 3 presents randomly generated samples from these models, demonstrating the progressive improvements in output sharpness.

A. Baseline VAE

The baseline convolutional model successfully captures cellular content, enabling both the reconstruction and generation of Ki67-stained histological images. However, the outputs exhibit reduced sharpness compared to real samples, which is a known limitation of standard VAE architectures. The average sharpness of this model is 51.37, which is approximately two times lower than the sharpness of real data. Fig. 2 (Model 1) illustrates the model's reconstruction performance, and Fig. 3 (Model 1) presents samples randomly generated from the latent space.

B. VAE with Alternative Reconstruction Losses

Results in Table I demonstrate that the choice of reconstruction loss function has a moderate influence on the sharpness of the generated outputs. Although the differences in sharpness are not substantial, the model trained with MAE achieves the

TABLE I
RESULTS OF EXPERIMENTS

	Baseline model	Experiment with reconstruction losses		Experiment with latent space dimensionalities			Experiment with sharpness aware objective function		
	Model configurations								
Reconstruction loss	**MSE**	BCE	**MAE**	MAE	MAE	MAE	MAE	MAE	MAE
Laten space dimensionalities	**128**	128	128	256	**512**	1024	512	512	512
Sharpness regularisation λ	**0**	0	0	0	0	0	1	**2**	3
	Results								
Sharpness (max)	**51.37**	54.52	**56.2**	68.83	**74.97**	75.01	75.99	**77.67**	79.37
MAE (min)	**0.2324**	0.2261	**0.2222**	0.1805	**0.1521**	0.1466	0.1536	**0.1513**	0.1516
KL divergence (min)	**186.51**	255.18	288.2	474.09	669.42	717.03	660.04	680.59	679.8

highest average sharpness, slightly outperforming the other loss functions. The performance advantage of MAE can be attributed to fundamental differences in error penalisation, as illustrated in Fig. 1. MSE employs a quadratic penalty (L_2 norm), which weights errors disproportionately, highlighting larger deviations with stronger penalties. Conversely, MAE utilises a linear penalty (L_1 norm), which treats errors uniformly. This characteristic allows MAE to better preserve edges and local details, thereby producing sharper outputs. Finally, BCE treats each pixel as an independent binary prediction and primarily penalises incorrect predictions at extreme pixel values (near 0 or 1); however, this characteristic provides only marginal improvement in sharpness for our histopathological image generation task. Based on these results, the MAE is selected for subsequent experiments as the most suitable reconstruction loss in terms of output sharpness. Fig. 2 (Model 2) illustrates images reconstructed by the MAE model, while Fig. 3 (Model 2) presents randomly generated samples from this model.

C. VAE with Alternative Latent Space Dimensionalities

Based on the results in Table I, the model with a latent dimensionality of 512 generates substantially sharper images than models with lower dimensionalities. These results confirm that a larger latent space improves output sharpness, as it allows the model to better capture salient features of the input data. However, we observe diminishing improvement with increasing dimensionality: expanding from 128 to 256 dimensions yields the largest sharpness gain across all experiments (12.63), followed by a more modest but still significant improvement of 6.14 when moving from 256 to 512, and a negligible gain of just 0.04 from 512 to 1024 dimensions. Given this diminishing improvement and the increase in computational complexity associated with higher dimensions, we identify 512 dimensions as the optimal trade-off between model performance and computational efficiency. Consequently, we adopt a latent dimensionality of 512 for all subsequent experiments. Fig. 2 (Model 3) illustrates the reconstruction performance of the 512-dimensional model, while Fig. 3 (Model 3) presents randomly generated samples. Both figures demonstrate a substantial increase in image sharpness, validating the efficacy of expanding the latent dimension.

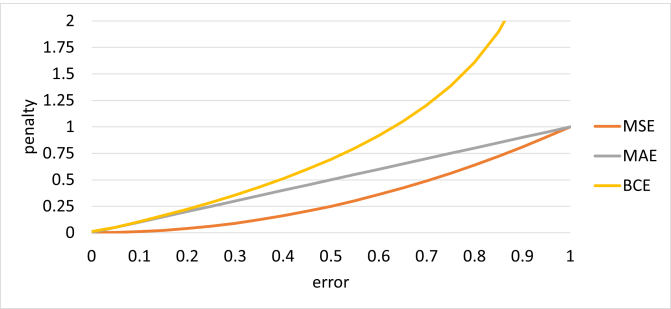

Fig. 1. Error penalisation curves for MSE, MAE, and BCE reconstruction loss functions. The x-axis represents the absolute pixel deviation between reconstructed and ground truth values (ranging from 0 to 1), and the y-axis shows the corresponding penalty.

D. VAE with Sharpness-Aware Objective Function

Table I presents the results for varying weighting coefficients, λ, indicating that incorporating the sharpness term into the objective function yields modest improvements in output sharpness. At $\lambda = 1$, the model achieves a marginal improvement of 1.02 over the baseline ($\lambda = 0$). Increasing λ to 2 yields a slightly larger gain of 1.68, and further increasing to $\lambda = 3$ produces a comparable improvement of 1.70. Notably, the relationship between λ, and reconstruction error is non-monotonic; while $\lambda = 1$ exhibits the highest MAE, $\lambda = 3$ also incurs a higher MAE than $\lambda = 2$ configuration. These results suggest that excessively high values of λ risk over-prioritising sharpness at the expense of reconstruction quality and KL divergence, both of which are critical for maintaining image structure and generation quality. Consequently, $\lambda = 2$ was selected as the optimal configuration for Ki67-stained image synthesis, as it provides sharpness improvement while accurately reconstructing and generating histological images. Fig. 2 (Model 4) illustrates the reconstruction performance at this setting, while Fig. 3 (Model 4) displays randomly generated samples.

E. Discussion

We evaluated three complementary approaches to improve the quality of generated Ki67-stained histological images, with particular emphasis on enhancing output sharpness.

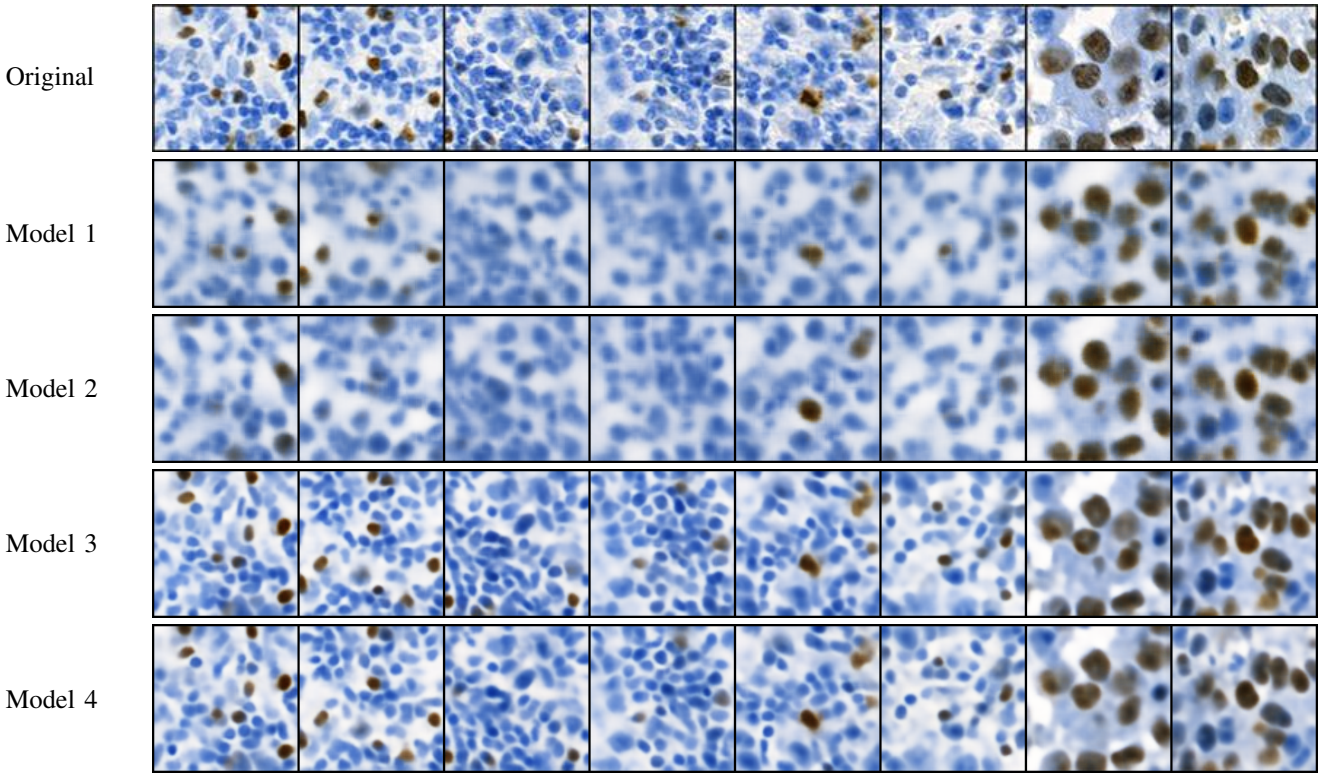

Fig. 2. Examples of image reconstructions by four VAE models. Top row: original Ki67-stained images. Subsequent rows: reconstructions by the baseline model (Model 1), MAE-based model (Model 2), 512-dimensional latent space model (Model 3), and sharpness-regularised model ($\lambda = 2$) (Model 4).

The first experiment compared three reconstruction loss functions: MSE, MAE, and BCE. The MAE-based model achieved the highest sharpness due to its linear L_1 penalty, which better preserves edges and fine details compared to the quadratic (L_2) penalty of MSE. Based on these results, MAE was adopted for all subsequent experiments.

The second experiment investigated the impact of latent space size on output quality. Notably, latent space dimensionality had the most substantial impact on output sharpness compared to the other investigated approaches. Increasing the latent dimensionality from 128 to 512 enabled the model to capture more detailed features of the input images, resulting in improved sharpness. However, we observed diminishing improvements beyond 512 dimensions, leading us to select this value as an optimal balance between performance and computational efficiency.

The third experiment incorporated a sharpness regularisation term based on the Sobel operator directly into the objective function. With an appropriately chosen weight ($\lambda = 2$), this modification yielded a modest but measurable improvement in output sharpness without compromising reconstruction quality.

Based on these findings, our final convolutional VAE model configuration consists of:

- MAE reconstruction loss,
- latent space dimensionality of 512,
- sharpness regularisation with $\lambda = 2$.

The model's reconstruction and generation capabilities are demonstrated in Fig. 2 and 3 (Model 4), respectively. However, despite these improvements, the final model achieved a sharpness score of 77.67, which remains substantially lower than the average sharpness of 103.76 observed in the real data. Visual inspection confirms that the generated images still exhibit noticeable blurring compared to real histological samples, indicating that further architectural modifications or alternative approaches may be necessary.

V. Conclusion

This work investigated the applicability of VAE for generating histological images, specifically Ki67-stained tissue samples. However, our dataset comprises exclusively seminoma tissue from a single institution, limiting generalisation to other tumour types, staining protocols, and patient populations. Through several experiments, we explored various modifications of the standard VAE aimed at improving the quality and sharpness of generated outputs.

The baseline convolutional VAE successfully captured cellular content characteristic of Ki67-stained tissues. However, the generated images consistently exhibited reduced sharpness compared to real samples, highlighting an inherent limitation of the VAE architecture. To address this limitation, we conducted three complementary experiments. First, we compared alternative reconstruction loss functions (MSE, MAE, BCE), finding that MAE provided the best preservation of finer details. Second, we investigated the impact of latent space dimensionality, determining that 512 dimensions offered the optimal balance between model capacity and computational

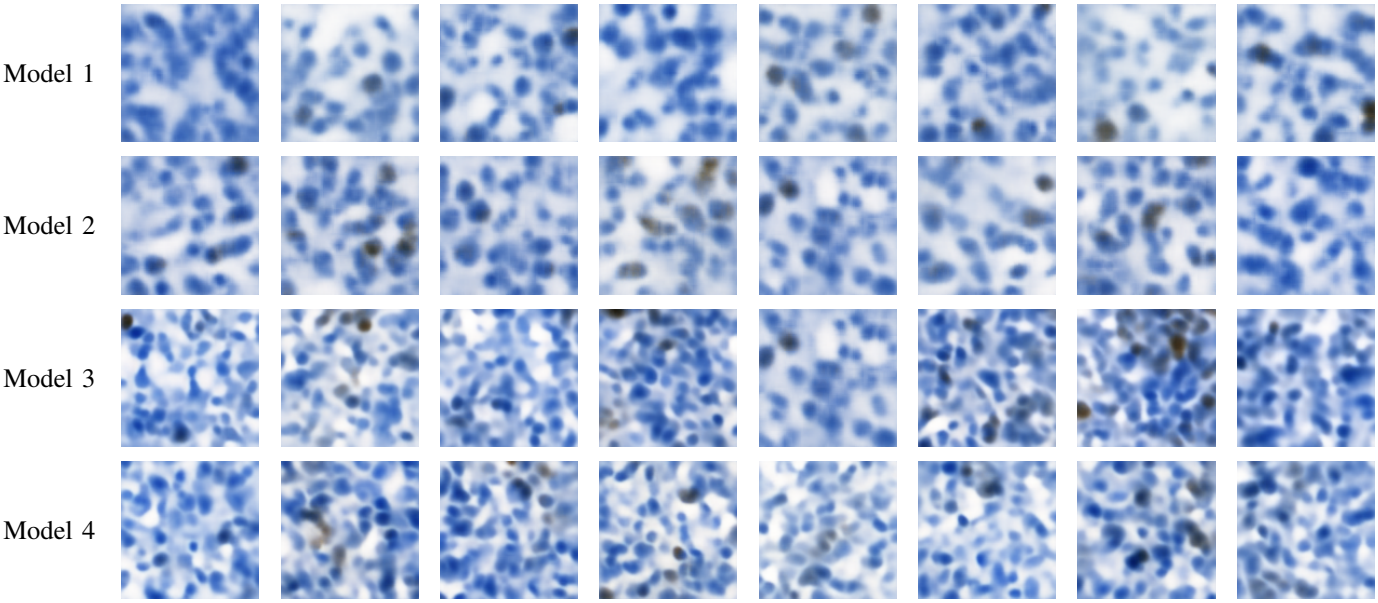

Fig. 3. Examples of generated Ki67-stained images by four VAE models: baseline model (Model 1), MAE-based model (Model 2), 512-dimensional latent space model (Model 3), and sharpness-regularised model ($\lambda = 2$) (Model 4).

efficiency. Third, we incorporated a sharpness-aware regularisation term into the objective function, yielding modest additional improvements. The final model, combining these modifications, achieved a sharpness score of 77.67 compared to 103.76 for real images, which is a substantial improvement over the baseline (sharpness of 51.37) but still indicates noticeable blurring in the outputs.

These findings indicate the potential of VAEs for medical image generation, particularly for data augmentation in scenarios with limited training samples or for other VAE-based histopathological frameworks. However, our results also reveal fundamental limitations of standard VAE architectures, primarily due to the smoothing effect inherent to the probabilistic reconstruction process. Several promising approaches exist for addressing these limitations, though they extend beyond standard VAE architectures. Hybrid architectures combining VAEs with GANs (VAE-GAN [3] or AVAE-S [8]) could produce sharper outputs. Alternative generative models, such as diffusion models, may also warrant investigation. Additionally, post-processing techniques could enhance the sharpness of VAE-generated images without modifying the core VAE architecture.

Acknowledgment

Thanks to L. Plank and K. Tobiášová for preparing the original dataset.

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